

Jun 23, 2022

Introduction to Computer and Network Security

Part 1

Question 1 [40 points]

In the context of cryptography, (a) describe the Kerckhoffs principle, (b) explain the purpose of key management, (c) explain the notion of symmetric key cryptography, and (d) explain the notion of asymmetric (or public) key cryptography.

Question 2 [10 points]

Describe an SQL injection attack. Which are the main mitigations to an SQL injection attack?

Part 2

Question 3 [40 points]

In the context of the Diffie-Hellman key exchange, describe (a) the protocol, (b) the Men-In-the-Middle (MITM) attack, and (c) how it is possible to mitigate the MITM attack.

Question 4 [10 points]

Define the notions of risk and vulnerability and give (at least) an example for each one.

Please, remember...

On each sheet, make sure to write:

Name

Surname

Student ID number

Question number